

# A 50 year old conjecture, disproven **change title**

Johan Voskamp

July 7, 2026

## Abstract

temp

## 1 AI/Data usage

## 2 Introduction

## 3 Prerequisites

To understand parts of the disproof of the - Theorem, we will have to define and explain some fundamental concepts beforehand.

### 3.1 Cayley Graphs

### 3.2 Minimal Cayley Graphs

A Cayley Graph is minimal if removing any element from the set that generates it, does not generate that cayley graph anymore. (So the generating set is minimal with respect to the Cayley graph). The minimal cayley graph has the least amount of edges while staying a representation of a group generated by S.

In a minimal Cayley graph, no cycle can contain a color exactly once, this is forced by the minimality: So basically we assume that we have a minimal cayley graph, and then we assume that we have a lonely edge in a cycle in that graph. Then by showing that that cycle can be rewritten as a product that expresses the lonely edge's generator as a product of the other generators, we see that that generator is redundant and hence the cayley graph was not minimal, giving us a contradiction.

$$\langle S \setminus \{s\} \rangle = \langle S \rangle = \Gamma(G, S)$$

Hence a lonely edge can't appear in a minimal cayley graph.

[proper proof necessary - Babai 1978]

### 3.3 No-lonely-colored graphs

It is quite evident that since all Cayley graphs are no-lonely-colored graphs, that Cayley Graphs are a subset of the broader spectrum of no-lonely-colored graphs.

[give example of a no-lonely-colored graph that is not Cayley here]

So if we find/conjecture anything that holds for all Cayley graphs, maybe then it will then also hold for all no-lonely-colored graphs.

### **3.4 The main conjecture**

No-lonely-colored graphs have a bounded chromatic number (Babai 1978)

### **3.5 Temp**

**4 a**

**5 b**